

Linear Data Structures: Stack

سٹیک LIFO پر چلتا ہے: جو سب سے بعد میں آیا وہ سب سے پہلے نکلے گا۔

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 17 of 24

★ Golden

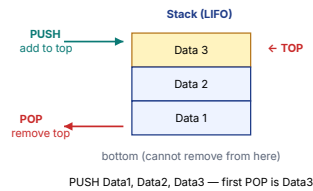
Linear Data Structures: Stack ★ Golden

LEARN IT

A **stack** is a linear data structure that works on the **LIFO** principle: **Last-In, First-Out**. The last element added is the first one removed. You cannot pull an item from the bottom or the middle.

- **PUSH** — add an item to the **top**.
- **POP** — remove the item from the **top**.

A **TOP** pointer always names the current top item. **Real-life example:** Undo in a text editor. Every action is piled on the stack; Undo POPs the most recent action first.



Three PUSH operations. The TOP is Data 3, so the first POP returns Data 3.

WORKED EXAMPLE

Worked example: PUSH / POP trace. Start with an empty stack. Then PUSH Data1, PUSH Data2, PUSH Data3, POP, POP.

Step	Operation	Stack (bottom → top)	TOP	Popped value
0	(empty)	[]	none	
1	PUSH Data1	[Data1]	Data1	
2	PUSH Data2	[Data1, Data2]	Data2	
3	PUSH Data3	[Data1, Data2, Data3]	Data3	
4	POP	[Data1, Data2]	Data2	Data3
5	POP	[Data1]	Data1	Data2

```
stack = []
stack.append("Data1")
stack.append("Data2")
stack.append("Data3")
print("after 3 PUSHes:", stack)
print("POP:", stack.pop())
print("POP:", stack.pop())
print("left:", stack)
```

```
OUTPUT
after 3 PUSHes: ['Data1', 'Data2', 'Data3']
POP: Data3
POP: Data2
left: ['Data1']
```

Python's `list.append` is PUSH and `list.pop()` (no index) is POP from the top.

REMEMBER IT

LIFO = Last In First Out = "last plate washed is first plate used" on a canteen pile. Undo, Redo, Back button and function calls are all stacks.

COMMON MISTAKE

Mistake: POPing from the bottom, or mixing LIFO with FIFO.

Fix: stack = plates (top only). Queue = ticket line (front leaves, rear joins). If the first POP is Data1, you drew a queue, not a stack.

اردو میں سمجھیں

سٹیک LIFO پر چلتا ہے: جو سب سے بعد میں آیا وہ سب سے پہلے نکلے گا۔ اوپر رکھنا PUSH، اوپر سے ہٹانا POP، اور TOP پوائنٹر موجودہ سب سے اوپر عنصر بتاتا ہے۔ انڈو (Undo) کی مثال: ہر عمل ڈھیر پر پڑتا ہے، انڈو سب سے تازہ عمل ہٹاتا ہے۔

EXAM TIP

Define stack + LIFO, define PUSH and POP, mention TOP, give the Undo example, and draw the vertical boxes with TOP labelled. Trace questions: "Show the stack after PUSH A, PUSH B, POP, PUSH C." Work left to right; the first POP must be B. Never remove from the middle.

CHECK YOURSELF

1. Expand LIFO.
2. After PUSH Data1, Data2, Data3, what does the first POP return?
3. Why is Undo modelled as a stack?

Answers: 1. Last-In, First-Out. 2. Data3. 3. Each action is PUSHed; Undo POPs the most recent action first.

Next lecture: Linear Data Structures: Queue